Naive Bayes

Machine Learning Techniques

Karthik Thiagarajan

Generative vs Discriminative

$$P(\mathbf{x}, y) = P(\mathbf{x}) \cdot P(y \mid \mathbf{x}) = P(y) \cdot P(\mathbf{x} \mid y)$$

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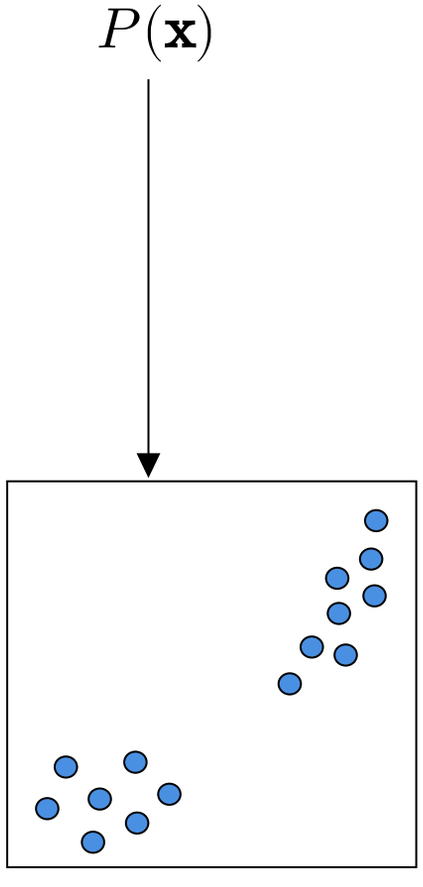

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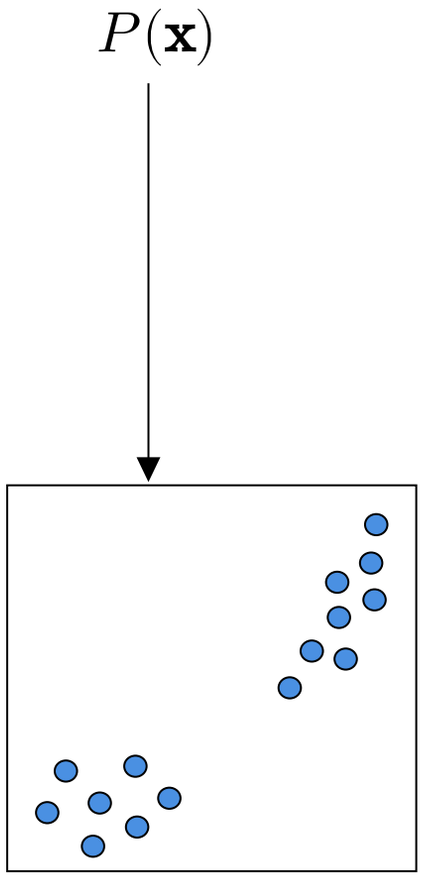

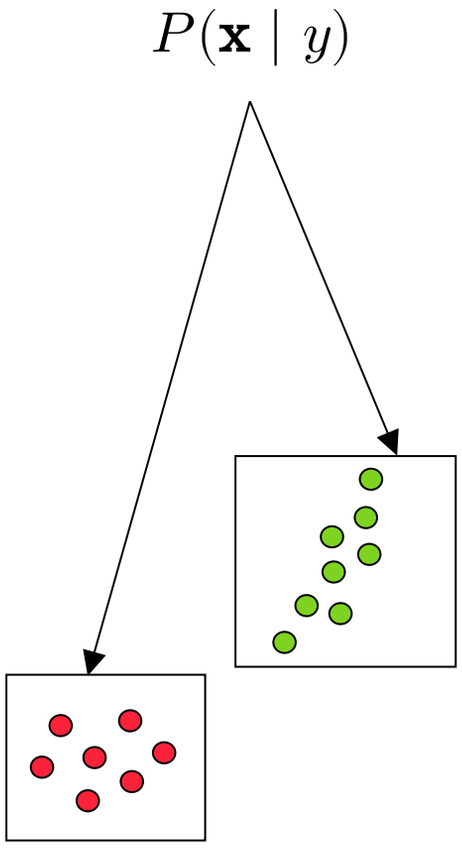

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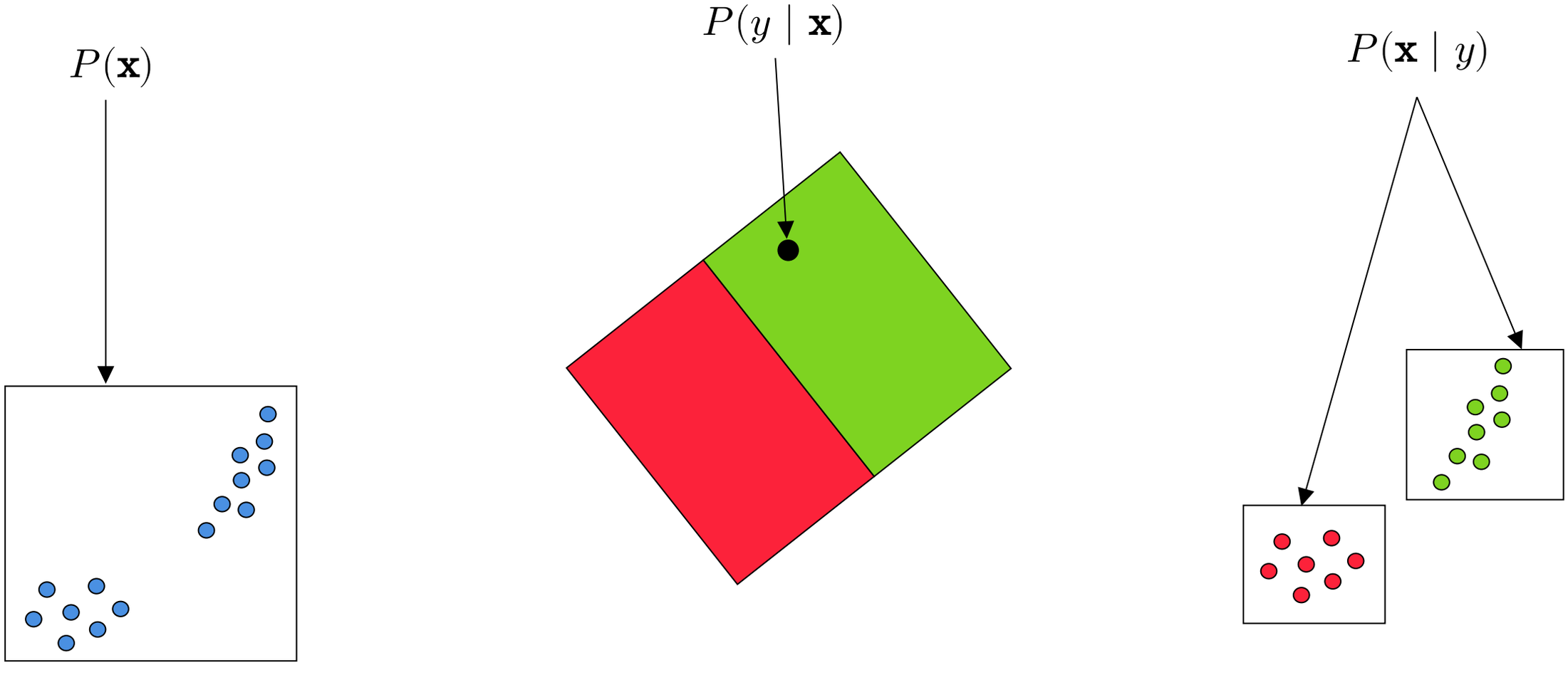

Text Classification

Text Classification

$\mathbf{x}$	y
two four eight six seven	0
six zero one four two	0
eight four zero six one three	0
seven one three	1
nine one three five seven	1
one three five six nine	1

Text Classification

$\mathbf{x}$	y
two four eight six seven	0
six zero one four two	0
eight four zero six one three	0
seven one three	1
nine one three five seven	1
one three five six nine	1

$$
\begin{aligned}
f_1 &\rightarrow \text{one} \\
&\vdots \\
f_9 &\rightarrow \text{nine} \\
f_{10} &\rightarrow \text{zero}
\end{aligned}
$$

Text Classification

$\mathbf{x}$	y
two four eight six seven	0
six zero one four two	0
eight four zero six one three	0
seven one three	1
nine one three five seven	1
one three five six nine	1

$$
\begin{aligned}
f_1 &\rightarrow \text{one} \\
&\vdots \\
f_9 &\rightarrow \text{nine} \\
f_{10} &\rightarrow \text{zero}
\end{aligned}
$$

$$
\text{seven one three} \rightarrow \begin{bmatrix} 1 \\ 0 \\ 1 \\ 0 \\ 0 \\ 0 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}
$$

Text Classification

$\mathbf{x}$	y
two four eight six seven	0
six zero one four two	0
eight four zero six one three	0
seven one three	1
nine one three five seven	1
one three five six nine	1

$$\begin{aligned} f_1 &\rightarrow \text{one} \\ &\vdots \\ f_9 &\rightarrow \text{nine} \\ f_{10} &\rightarrow \text{zero} \end{aligned}$$

$$\mathbf{x} = \begin{bmatrix} f_1 \\ \vdots \\ f_{10} \end{bmatrix}$$

$$\text{seven one three} \rightarrow \begin{bmatrix} 1 \\ 0 \\ 1 \\ 0 \\ 0 \\ 0 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

Text Classification

$\mathbf{x}$	y
two four eight six seven	0
six zero one four two	0
eight four zero six one three	0
seven one three	1
nine one three five seven	1
one three five six nine	1

$$\mathbf{x} \in \mathbb{R}^{10} \qquad y \in \{0, 1\}$$

$$\begin{aligned} f_1 &\rightarrow \text{one} \\ &\vdots \\ f_9 &\rightarrow \text{nine} \\ f_{10} &\rightarrow \text{zero} \end{aligned}$$

$$\mathbf{x} = \begin{bmatrix} f_1 \\ \vdots \\ f_{10} \end{bmatrix}$$

$$\text{seven one three} \rightarrow \begin{bmatrix} 1 \\ 0 \\ 1 \\ 0 \\ 0 \\ 0 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

Text Classification

$$(\mathbf{x}, y) = \big(\text{one five eight},\ 1\big)$$

Text Classification

$(\mathbf{x}, y) = \big(\text{one five eight},\ 1\big)$

$\boxed{y = 1}$ one two three four five six seven eight nine zero

$\boxed{y = 0}$ one two three four five six seven eight nine zero

Text Classification

$(\mathbf{x}, y) = \big(\text{one five eight},\ 1\big)$

$y = 1$ one two three four five six seven eight nine zero

$y = 0$ one two three four five six seven eight nine zero

Text Classification

$(\mathbf{x}, y) = \big(\text{one five eight},\ 1\big)$

$y = 1$ one two three four five six seven eight nine zero

one five eight

$y = 0$ one two three four five six seven eight nine zero

Class Conditional Independence

$$P(\mathbf{x} \mid y) = P(x_1, \cdots, x_d \mid y)$$

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$$P\Big(\text{seven one three} \mid 1\Big) = P\Big(\text{seven} \mid 1\Big) \cdot P\Big(\text{one} \mid 1\Big) \cdot P\Big(\text{three} \mid 1\Big)$$

Class Conditional Independence

$$P(\mathbf{x} \mid y) = P(x_1, \cdots, x_d \mid y) = P(x_1 \mid y) \cdots P(x_d \mid y)$$

$$\begin{aligned} P\Big(\text{seven one three} \mid 1\Big) = {} & P\Big(\text{seven} \mid 1\Big) \cdot P\Big(\text{one} \mid 1\Big) \cdot P\Big(\text{three} \mid 1\Big) \\ & P\Big(\text{tw}\cancel{\text{o}} \mid 1\Big) \cdot P\Big(\text{fo}\cancel{\text{ur}} \mid 1\Big) \cdot P\Big(\text{fi}\cancel{\text{ve}} \mid 1\Big) \\ & P\Big(\text{s}\cancel{\text{ix}} \mid 1\Big) \cdot P\Big(\text{ei}\cancel{\text{ght}} \mid 1\Big) \cdot P\Big(\text{ni}\cancel{\text{ne}} \mid 1\Big) \\ & P\Big(\text{ze}\cancel{\text{ro}} \mid 1\Big) \end{aligned}$$

~~Class Conditional Independence~~

$$\# \text{ of parameters} =$$

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$$\#\text{ of parameters} = 2 \cdot 1023 + 1 = 2047$$

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$$\# \text{ of parameters} = 2 \cdot 1023 + 1 = 2047$$

$$\underbrace{2}_{\text{Number of classes}} \cdot \underbrace{\left(2^d - 1\right)}_{\substack{\text{Number of parameters} \\ \text{for one class}}} + \underbrace{1}_{\substack{\text{Number of parameters} \\ \text{for prior}}}$$

Class Conditional Independence

$$\# \text{ of parameters} =$$

Class Conditional Independence

$$\#\text{ of parameters} = 2 \cdot 10 + 1 = 21$$

Class Conditional Independence

$$\text{\# of parameters} = 2 \cdot 10 + 1 = 21$$

$$\underbrace{2}_{\text{Number of classes}} \cdot \underbrace{d}_{\substack{\text{Number of parameters} \\ \text{for one class}}} + \underbrace{1}_{\substack{\text{Number of parameters} \\ \text{for prior}}}$$

MLE

- $p \quad \rightarrow$ probability of picking class 1
- $x_{ij} \quad \rightarrow$ presence of jth word in i^{th} sentence
- $p_j^y \quad \rightarrow$ probability of seeing j^{th} word in class y

MLE

$$L(\theta; D) = \prod_{i=1}^{n} p^{y_i} \cdot (1-p)^{1-y_i} \cdot \prod_{j=1}^{d} \left(p_j^{y_i}\right)^{x_{ij}} \cdot \left(1-p_j^{y_i}\right)^{1-x_{ij}}$$

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MLE

$$L(\theta; D) = \prod_{i=1}^{n} p^{y_i} \cdot (1-p)^{1-y_i} \cdot \prod_{j=1}^{d} \left(p_j^{y_i}\right)^{x_{ij}} \cdot \left(1-p_j^{y_i}\right)^{1-x_{ij}}$$

$$l(\theta; D) = \sum_{i=1}^{n} [y_i \log p + (1-y_i)\log(1-p)] +$$

$$\sum_{i=1}^{n} \sum_{j=1}^{d} \left[x_{ij} \cdot \log\left(p_j^{y_i}\right) + (1-x_{ij}) \cdot \log\left(1-p_j^{y_i}\right) \right]$$

MLE

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$$\frac{\partial l}{\partial p_j^y} = 0$$

MLE

$$L(\theta; D) = \prod_{i=1}^{n} p^{y_i} \cdot (1-p)^{1-y_i} \cdot \prod_{j=1}^{d} \left(p_j^{y_i}\right)^{x_{ij}} \cdot \left(1-p_j^{y_i}\right)^{1-x_{ij}}$$

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$$\frac{\partial l}{\partial p_j^y} = 0$$

$$\sum_{i=1}^{n} \mathbb{I}[y_i = y] \left[\frac{x_{ij}}{p_j^y} - (1-x_{ij}) \cdot \frac{1}{1-p_j^y} \right] = 0$$

MLE

$$L(\theta; D) = \prod_{i=1}^{n} p^{y_i} \cdot (1-p)^{1-y_i} \cdot \prod_{j=1}^{d} \left(p_j^{y_i}\right)^{x_{ij}} \cdot \left(1-p_j^{y_i}\right)^{1-x_{ij}}$$

$$l(\theta; D) = \sum_{i=1}^{n} [y_i \log p + (1-y_i)\log(1-p)] +$$

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$$\sum_{i=1}^{n} \mathbb{I}[y_i = y] \left[x_{ij}\left(1-p_j^y\right) - (1-x_{ij}) \cdot p_j^y \right] = 0$$

$$\sum_{i=1}^{n} \mathbb{I}[y_i = y] \left[x_{ij} - p_j^y \right] = 0$$

MLE

$$L(\theta; D) = \prod_{i=1}^{n} p^{y_i} \cdot (1-p)^{1-y_i} \cdot \prod_{j=1}^{d} \left(p_j^{y_i}\right)^{x_{ij}} \cdot \left(1-p_j^{y_i}\right)^{1-x_{ij}}$$

$$l(\theta; D) = \sum_{i=1}^{n} [y_i \log p + (1-y_i)\log(1-p)] +$$

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$$\sum_{i=1}^{n} \mathbb{I}[y_i = y] \left[\frac{x_{ij}}{p_j^y} - (1-x_{ij}) \cdot \frac{1}{1-p_j^y} \right] = 0$$

$$\sum_{i=1}^{n} \mathbb{I}[y_i = y] \left[x_{ij}\left(1-p_j^y\right) - (1-x_{ij}) \cdot p_j^y \right] = 0$$

$$\sum_{i=1}^{n} \mathbb{I}[y_i = y] \left[x_{ij} - p_j^y \right] = 0$$

$$\boxed{p_j^y = \frac{\sum_{i=1}^{n} x_{ij} \cdot \mathbb{I}[y_i = y]}{\sum_{i=1}^{n} \mathbb{I}[y_i = y]}}$$

MLE

$$p_j^y = \frac{\sum_{i=1}^{n} x_{ij} \cdot \mathbb{I}[y_i = y]}{\sum_{i=1}^{n} \mathbb{I}[y_i = y]}$$

$$p = \frac{\sum_{i=1}^{n} \mathbb{I}[y_i = 1]}{n}$$

Naive Bayes

$\mathbf{x}$	y
two four eight six seven	0
six zero one four two	0
eight four zero six one three	0
seven one three	1
nine one three five seven	1
one three five six nine	1

Naive Bayes

$\mathbf{x}$	y
two four eight six seven	0
six zero one four two	0
eight four zero six one three	0
seven one three	1
nine one three five seven	1
one three five six nine	1

$$p =$$

Naive Bayes

$\mathbf{x}$	y
two four eight six seven	0
six zero one four two	0
eight four zero six one three	0
seven one three	1
nine one three five seven	1
one three five six nine	1

$$p = \frac{1}{2}$$

j	p_j^0
1	
2	
3	
4	
5	
6	
7	
8	
9	
0	

j	p_j^1
1	
2	
3	
4	
5	
6	
7	
8	
9	
0	

Naive Bayes

$\mathbf{x}$	y
two four eight six seven	0
six zero one four two	0
eight four zero six one three	0
seven one three	1
nine one three five seven	1
one three five six nine	1

$$p = \frac{1}{2}$$

j	p_j^0
1	2 / 3
2	2 / 3
3	1 / 3
4	1
5	0
6	
7	
8	
9	
0	

j	p_j^1
1	
2	
3	
4	
5	
6	
7	
8	
9	
0	

Naive Bayes

$\mathbf{x}$	y
two four eight six seven	0
six zero one four two	0
eight four zero six one three	0
seven one three	1
nine one three five seven	1
one three five six nine	1

$$p = \frac{1}{2}$$

j	p_j^0
1	2 / 3
2	2 / 3
3	1 / 3
4	1
5	0
6	1
7	1 / 3
8	2 / 3
9	0
0	2 / 3

j	p_j^1
1	
2	
3	
4	
5	
6	
7	
8	
9	
0	

Naive Bayes

$\mathbf{x}$	y
two four eight six seven	0
six zero one four two	0
eight four zero six one three	0
seven one three	1
nine one three five seven	1
one three five six nine	1

$$p = \frac{1}{2}$$

j	p_j^0
1	2 / 3
2	2 / 3
3	1 / 3
4	1
5	0
6	1
7	1 / 3
8	2 / 3
9	0
0	2 / 3

j	p_j^1
1	1
2	0
3	1
4	0
5	2 / 3
6	1 / 3
7	2 / 3
8	0
9	2 / 3
0	0

Naive Bayes

j	p_j^0
1	$2/3$
2	$2/3$
3	$1/3$
4	1
5	0
6	1
7	$1/3$
8	$2/3$
9	0
0	$2/3$

j	p_j^1
1	1
2	0
3	1
4	0
5	$2/3$
6	$1/3$
7	$2/3$
8	0
9	$2/3$
0	0

$$P\big(y = 1 \mid \text{eight two one six}\big)$$

Naive Bayes

j	p_j^0
1	$2/3$
2	$2/3$
3	$1/3$
4	1
5	0
6	1
7	$1/3$
8	$2/3$
9	0
0	$2/3$

j	p_j^1
1	1
2	0
3	1
4	0
5	$2/3$
6	$1/3$
7	$2/3$
8	0
9	$2/3$
0	0

$$P\Big(y = 1 \mid \text{eight two one six}\Big)$$

$$= \frac{P(y=1) \cdot P\Big(\text{eight two one six} \mid y \;=\; 1\Big)}{P\Big(\text{eight two one six}\Big)}$$

Naive Bayes

j	p_j^0
1	$2/3$
2	$2/3$
3	$1/3$
4	1
5	0
6	1
7	$1/3$
8	$2/3$
9	0
0	$2/3$

j	p_j^1
1	1
2	0
3	1
4	0
5	$2/3$
6	$1/3$
7	$2/3$
8	0
9	$2/3$
0	0

$$P\big(\text{eight two one six} \mid y = 1\big)$$

Naive Bayes

j	p_j^0
1	$2/3$
2	$2/3$
3	$1/3$
4	1
5	0
6	1
7	$1/3$
8	$2/3$
9	0
0	$2/3$

j	p_j^1
1	1
2	0
3	1
4	0
5	$2/3$
6	$1/3$
7	$2/3$
8	0
9	$2/3$
0	0

$$P\big(\text{eight two one six} \mid y = 1\big)$$

$$= P\Big(\big[1 \;\; 1 \;\; 0 \;\; 0 \;\; 0 \;\; 1 \;\; 0 \;\; 1 \;\; 0 \;\; 0\big]^T \mid y = 1\Big)$$

Naive Bayes

j	p_j^0
1	$2/3$
2	$2/3$
3	$1/3$
4	1
5	0
6	1
7	$1/3$
8	$2/3$
9	0
0	$2/3$

j	p_j^1
1	1
2	0
3	1
4	0
5	$2/3$
6	$1/3$
7	$2/3$
8	0
9	$2/3$
0	0

$$P\big(\text{eight two one six} \mid y = 1\big)$$

$$= P\Big(\big[1 \;\; 1 \;\; 0 \;\; 0 \;\; 0 \;\; 1 \;\; 0 \;\; 1 \;\; 0 \;\; 0\big]^T \mid y = 1\Big)$$

$$= p_1^1 \cdot p_2^1 \cdot$$

$$p_6^1 \cdot \qquad\qquad \cdot p_8^1 \cdot$$

Naive Bayes

j	p_j^0
1	$2/3$
2	$2/3$
3	$1/3$
4	1
5	0
6	1
7	$1/3$
8	$2/3$
9	0
0	$2/3$

j	p_j^1
1	1
2	0
3	1
4	0
5	$2/3$
6	$1/3$
7	$2/3$
8	0
9	$2/3$
0	0

$$P\Big(\text{eight two one six} \mid y = 1\Big)$$

$$= P\Big(\begin{bmatrix} 1 & 1 & 0 & 0 & 0 & 1 & 0 & 1 & 0 & 0 \end{bmatrix}^T \mid y = 1\Big)$$

$$= p_1^1 \cdot p_2^1 \cdot \left(1 - p_3^1\right) \cdot \left(1 - p_4^1\right) \cdot \left(1 - p_5^1\right)$$
$$p_6^1 \cdot \left(1 - p_7^1\right) \cdot p_8^1 \cdot \left(1 - p_9^1\right) \cdot \left(1 - p_{10}^1\right)$$

$$= 0$$

Naive Bayes

j	p_j^0
1	$2/3$
2	$2/3$
3	$1/3$
4	1
5	0
6	1
7	$1/3$
8	$2/3$
9	0
0	$2/3$

j	p_j^1
1	1
2	0
3	1
4	0
5	$2/3$
6	$1/3$
7	$2/3$
8	0
9	$2/3$
0	0

$$P\Big(\text{eight two one six} \mid y = 0\Big)$$

$$= P\Big(\begin{bmatrix} 1 & 1 & 0 & 0 & 0 & 1 & 0 & 1 & 0 & 0 \end{bmatrix}^T \mid y = 0\Big)$$

$$= p_1^0 \cdot p_2^0 \cdot \left(1 - p_3^0\right) \cdot \left(1 - p_4^0\right) \cdot \left(1 - p_5^0\right)$$
$$p_6^0 \cdot \left(1 - p_7^0\right) \cdot p_8^0 \cdot \left(1 - p_9^0\right) \cdot \left(1 - p_{10}^0\right)$$

Naive Bayes

j	p_j^0
1	$2/3$
2	$2/3$
3	$1/3$
4	1
5	0
6	1
7	$1/3$
8	$2/3$
9	0
0	$2/3$

j	p_j^1
1	1
2	0
3	1
4	0
5	$2/3$
6	$1/3$
7	$2/3$
8	0
9	$2/3$
0	0

$$P\Big(\text{eight two one six} \mid y = 0\Big)$$

$$= P\Big(\begin{bmatrix} 1 & 1 & 0 & 0 & 0 & 1 & 0 & 1 & 0 & 0 \end{bmatrix}^T \mid y = 0\Big)$$

$$= p_1^0 \cdot p_2^0 \cdot \left(1 - p_3^0\right) \cdot \left(1 - p_4^0\right) \cdot \left(1 - p_5^0\right)$$
$$p_6^0 \cdot \left(1 - p_7^0\right) \cdot p_8^0 \cdot \left(1 - p_9^0\right) \cdot \left(1 - p_{10}^0\right)$$

$$= 0$$

Laplace Smoothing

$\mathbf{x}$	y
two four eight six seven	0
six zero one four two	0
eight four zero six one three	0
one two three four five six seven eight nine zero	0
seven one three	1
nine one three five seven	1
one three five six nine	1
one two three four five six seven eight nine zero	1

$$p = \frac{1}{2}$$

Laplace Smoothing

$\mathbf{x}$	y
two four eight six seven	0
six zero one four two	0
eight four zero six one three	0
one two three four five six seven eight nine zero	0
seven one three	1
nine one three five seven	1
one three five six nine	1
one two three four five six seven eight nine zero	1

$$p = \frac{1}{2}$$

j	p_j^0
1	
2	
3	
4	
5	
6	
7	
8	
9	
0	

j	p_j^1
1	
2	
3	
4	
5	
6	
7	
8	
9	
0	

Laplace Smoothing

$\mathbf{x}$	y
two four eight six seven	0
six zero one four two	0
eight four zero six one three	0
one two three four five six seven eight nine zero	0
seven one three	1
nine one three five seven	1
one three five six nine	1
one two three four five six seven eight nine zero	1

$$p = \frac{1}{2}$$

j	p_j^0
1	$3/4$
2	$3/4$
3	$1/2$
4	1
5	$1/4$
6	1
7	$1/2$
8	$3/4$
9	$1/4$
0	$3/4$

j	p_j^1
1	1
2	$1/4$
3	1
4	$1/4$
5	$3/4$
6	$1/2$
7	$3/4$
8	$1/4$
9	$3/4$
0	$1/4$

Naive Bayes

j	p_j^0
1	$3/4$
2	$3/4$
3	$1/2$
4	1
5	$1/4$
6	1
7	$1/2$
8	$3/4$
9	$1/4$
0	$3/4$

j	p_j^1
1	1
2	$1/4$
3	1
4	$1/4$
5	$3/4$
6	$1/2$
7	$3/4$
8	$1/4$
9	$3/4$
0	$1/4$

$$P\Big(\text{eight two one six} \mid y = 0\Big)$$

$$= P\Big(\begin{bmatrix} 1 & 1 & 0 & 0 & 0 & 1 & 0 & 1 & 0 & 0 \end{bmatrix}^T \mid y = 0\Big)$$

$$= p_1^0 \cdot p_2^0 \cdot \left(1 - p_3^0\right) \cdot \left(1 - p_4^0\right) \cdot \left(1 - p_5^0\right)$$
$$p_6^0 \cdot \left(1 - p_7^0\right) \cdot p_8^0 \cdot \left(1 - p_9^0\right) \cdot \left(1 - p_{10}^0\right)$$

Naive Bayes

j	p_j^0
1	$3/4$
2	$3/4$
3	$1/2$
4	1
5	$1/4$
6	1
7	$1/2$
8	$3/4$
9	$1/4$
0	$3/4$

j	p_j^1
1	1
2	$1/4$
3	1
4	$1/4$
5	$3/4$
6	$1/2$
7	$3/4$
8	$1/4$
9	$3/4$
0	$1/4$

$$P\Big(\text{eight two one six} \mid y = 0\Big)$$

$$= P\Big(\begin{bmatrix} 1 & 1 & 0 & 0 & 0 & 1 & 0 & 1 & 0 & 0 \end{bmatrix}^T \mid y = 0\Big)$$

$$= p_1^0 \cdot p_2^0 \cdot \left(1 - p_3^0\right) \cdot \left(1 - p_4^0\right) \cdot \left(1 - p_5^0\right)$$
$$p_6^0 \cdot \left(1 - p_7^0\right) \cdot p_8^0 \cdot \left(1 - p_9^0\right) \cdot \left(1 - p_{10}^0\right)$$

$$= 0$$

Naive Bayes

j	p_j^0
1	$3/4$
2	$3/4$
3	$1/2$
4	1
5	$1/4$
6	1
7	$1/2$
8	$3/4$
9	$1/4$
0	$3/4$

j	p_j^1
1	1
2	$1/4$
3	1
4	$1/4$
5	$3/4$
6	$1/2$
7	$3/4$
8	$1/4$
9	$3/4$
0	$1/4$

$$P\Big(\text{eight two one six} \mid y = 1\Big)$$

$$= P\Big(\begin{bmatrix} 1 & 1 & 0 & 0 & 0 & 1 & 0 & 1 & 0 & 0 \end{bmatrix}^T \mid y = 1\Big)$$

$$= p_1^1 \cdot p_2^1 \cdot \left(1 - p_3^1\right) \cdot \left(1 - p_4^1\right) \cdot \left(1 - p_5^1\right)$$
$$p_6^1 \cdot \left(1 - p_7^1\right) \cdot p_8^1 \cdot \left(1 - p_9^1\right) \cdot \left(1 - p_{10}^1\right)$$

Naive Bayes

j	p_j^0
1	$3/4$
2	$3/4$
3	$1/2$
4	1
5	$1/4$
6	1
7	$1/2$
8	$3/4$
9	$1/4$
0	$3/4$

j	p_j^1
1	1
2	$1/4$
3	1
4	$1/4$
5	$3/4$
6	$1/2$
7	$3/4$
8	$1/4$
9	$3/4$
0	$1/4$

$$P\Big(\text{eight two one six} \mid y = 1\Big)$$

$$= P\Big(\begin{bmatrix} 1 & 1 & 0 & 0 & 0 & 1 & 0 & 1 & 0 & 0 \end{bmatrix}^T \mid y = 1\Big)$$

$$= p_1^1 \cdot p_2^1 \cdot \left(1 - p_3^1\right) \cdot \left(1 - p_4^1\right) \cdot \left(1 - p_5^1\right)$$
$$p_6^1 \cdot \left(1 - p_7^1\right) \cdot p_8^1 \cdot \left(1 - p_9^1\right) \cdot \left(1 - p_{10}^1\right)$$

$$= 0$$

Laplace Smoothing

$\mathbf{x}$	y
two four eight six seven	0
six zero one four two	0
eight four zero six one three	0
one two three four five six seven eight nine zero	0
	0
seven one three	1
nine one three five seven	1
one three five six nine	1
one two three four five six seven eight nine zero	1
	1

$$p = \frac{1}{2}$$

Laplace Smoothing

$\mathbf{x}$	y
two four eight six seven	0
six zero one four two	0
eight four zero six one three	0
one two three four five six seven eight nine zero	0
	0
seven one three	1
nine one three five seven	1
one three five six nine	1
one two three four five six seven eight nine zero	1
	1

j	p_j^0
1	$3/5$
2	$3/5$
3	$2/5$
4	$4/5$
5	$1/5$
6	$4/5$
7	$2/5$
8	$3/5$
9	$1/5$
0	$3/5$

j	p_j^1
1	$4/5$
2	$1/5$
3	$4/5$
4	$1/5$
5	$3/5$
6	$2/5$
7	$3/5$
8	$1/5$
9	$3/5$
0	$1/5$

$$p = \frac{1}{2}$$

Naive Bayes

j	p_j^0
1	$3/5$
2	$3/5$
3	$2/5$
4	$4/5$
5	$1/5$
6	$4/5$
7	$2/5$
8	$3/5$
9	$1/5$
0	$3/5$

j	p_j^1
1	$4/5$
2	$1/5$
3	$4/5$
4	$1/5$
5	$3/5$
6	$2/5$
7	$3/5$
8	$1/5$
9	$3/5$
0	$1/5$

$$P\Big(\text{eight two one six} \mid y = 0\Big)$$

$$= P\Big(\begin{bmatrix} 1 & 1 & 0 & 0 & 0 & 1 & 0 & 1 & 0 & 0 \end{bmatrix}^T \mid y = 0\Big)$$

$$= p_1^0 \cdot p_2^0 \cdot \left(1 - p_3^0\right) \cdot \left(1 - p_4^0\right) \cdot \left(1 - p_5^0\right)$$
$$p_6^0 \cdot \left(1 - p_7^0\right) \cdot p_8^0 \cdot \left(1 - p_9^0\right) \cdot \left(1 - p_{10}^0\right)$$

$$\approx 0.0032$$

Naive Bayes

j	p_j^0
1	$3/5$
2	$3/5$
3	$2/5$
4	$4/5$
5	$1/5$
6	$4/5$
7	$2/5$
8	$3/5$
9	$1/5$
0	$3/5$

j	p_j^1
1	$4/5$
2	$1/5$
3	$4/5$
4	$1/5$
5	$3/5$
6	$2/5$
7	$3/5$
8	$1/5$
9	$3/5$
0	$1/5$

$$P\Big(\text{eight two one six} \mid y = 1\Big)$$

$$= P\Big(\begin{bmatrix} 1 & 1 & 0 & 0 & 0 & 1 & 0 & 1 & 0 & 0 \end{bmatrix}^T \mid y = 1\Big)$$

$$= p_1^1 \cdot p_2^1 \cdot \left(1 - p_3^1\right) \cdot \left(1 - p_4^1\right) \cdot \left(1 - p_5^1\right)$$
$$p_6^1 \cdot \left(1 - p_7^1\right) \cdot p_8^1 \cdot \left(1 - p_9^1\right) \cdot \left(1 - p_{10}^1\right)$$

$$\approx 0.0001$$

Naive Bayes | Prediction

$$P\Big(y = 1 \mid \text{eight two one six}\Big) =$$

Naive Bayes | Prediction

$$P\Big(y = 1 \mid \text{eight two one six}\Big) = \frac{P(y = 1) \cdot P\Big(\text{eight two one six} \mid y = 1\Big)}{P\Big(\text{eight two one six}\Big)}$$

Naive Bayes | Prediction

$$P\Big(y = 1 \mid \text{eight two one six}\Big) = \frac{P(y=1) \cdot P\Big(\text{eight two one six} \mid y = 1\Big)}{P\Big(\text{eight two one six}\Big)}$$

$$= \frac{P(y=1) \cdot P\Big(\text{eight two one six} \mid y = 1\Big)}{P(y=1) \cdot P\Big(\text{eight two one six} \mid y = 1\Big) + P(y=0) \cdot P\Big(\text{eight two one six} \mid y = 0\Big)}$$

Naive Bayes | Prediction

$$P\Big(y = 1 \mid \text{eight two one six}\Big) = \frac{P(y=1) \cdot P\Big(\text{eight two one six} \mid y = 1\Big)}{P\Big(\text{eight two one six}\Big)}$$

$$= \frac{P(y=1) \cdot P\Big(\text{eight two one six} \mid y = 1\Big)}{P(y=1) \cdot P\Big(\text{eight two one six} \mid y = 1\Big) + P(y=0) \cdot P\Big(\text{eight two one six} \mid y = 0\Big)}$$

$$= \frac{0.5 \times 0.0001}{0.5 \times 0.0001 + 0.5 \times 0.0032}$$

$$= 0.03$$

Naive Bayes | Prediction

$$P\Big(y = 0 \mid \text{eight two one six}\Big) = \frac{P(y=0) \cdot P\Big(\text{eight two one six} \mid y = 0\Big)}{P\Big(\text{eight two one six}\Big)}$$

$$= \frac{P(y=0) \cdot P\Big(\text{eight two one six} \mid y = 0\Big)}{P(y=1) \cdot P\Big(\text{eight two one six} \mid y = 1\Big) + P(y=0) \cdot P\Big(\text{eight two one six} \mid y = 0\Big)}$$

$$= \frac{0.5 \times 0.0032}{0.5 \times 0.0001 + 0.5 \times 0.0032}$$

$$= 0.97$$

Naive Bayes | Prediction

$$P(y = 1 \mid \mathbf{x}) \propto P(y = 1) \cdot P(\mathbf{x} \mid y = 1)$$

$$P(y = 0 \mid \mathbf{x}) \propto P(y = 0) \cdot P(\mathbf{x} \mid y = 0)$$

Naive Bayes | Prediction

prior

posterior

$$P(y = 1 \mid \mathbf{x}) \propto P(y = 1) \cdot P(\mathbf{x} \mid y = 1)$$

likelihood

$$P(y = 0 \mid \mathbf{x}) \propto P(y = 0) \cdot P(\mathbf{x} \mid y = 0)$$

Naive Bayes | Prediction

prior

posterior $\longrightarrow$ $$P(y = 1 \mid \mathbf{x}) \propto P(y = 1) \cdot P(\mathbf{x} \mid y = 1)$$ $\longleftarrow$ likelihood

$$P(y = 0 \mid \mathbf{x}) \propto P(y = 0) \cdot P(\mathbf{x} \mid y = 0)$$

$$P(\mathbf{x} \mid y = 0) \gtrless P(\mathbf{x} \mid y = 1)$$ $\longleftarrow$ only for equal priors!

Naive Bayes | Decision Boundary

$$\frac{P(y=1 \mid \mathbf{x})}{P(y=0 \mid \mathbf{x})} = \frac{P(y=1)}{P(y=0)} \cdot \frac{P(\mathbf{x} \mid y=1)}{P(\mathbf{x} \mid y=0)}$$

$$= \frac{p}{1-p} \cdot \frac{\prod_{j=1}^{d} \left(p_j^1\right)^{x_j} \cdot \left(1-p_j^1\right)^{1-x_j}}{\prod_{j=1}^{d} \left(p_j^0\right)^{x_j} \cdot \left(1-p_j^0\right)^{1-x_j}}$$

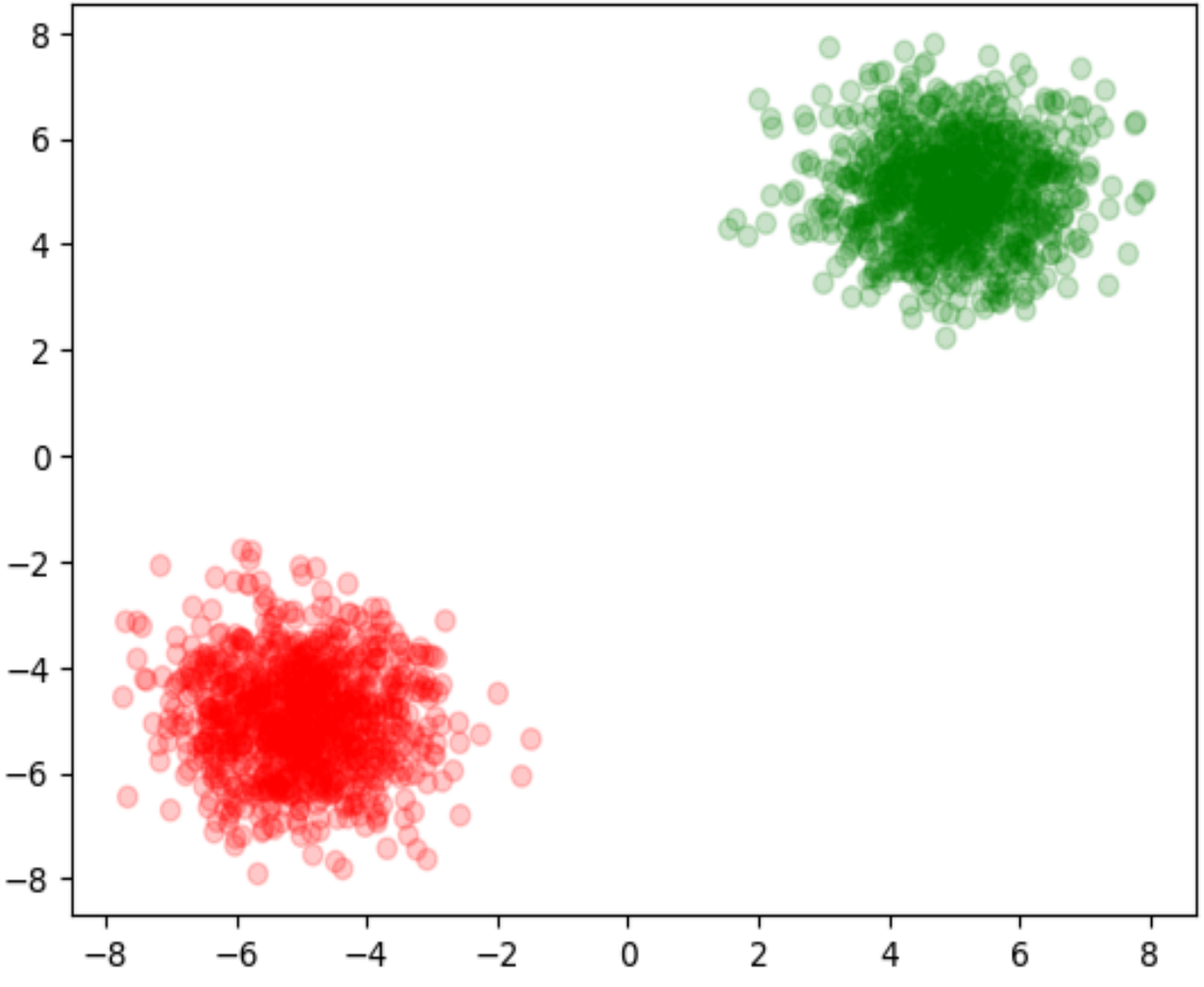
Naive Bayes | Decision Boundary

$$\log\left[\frac{P(y=1 \mid \mathbf{x})}{P(y=0 \mid \mathbf{x})}\right]$$

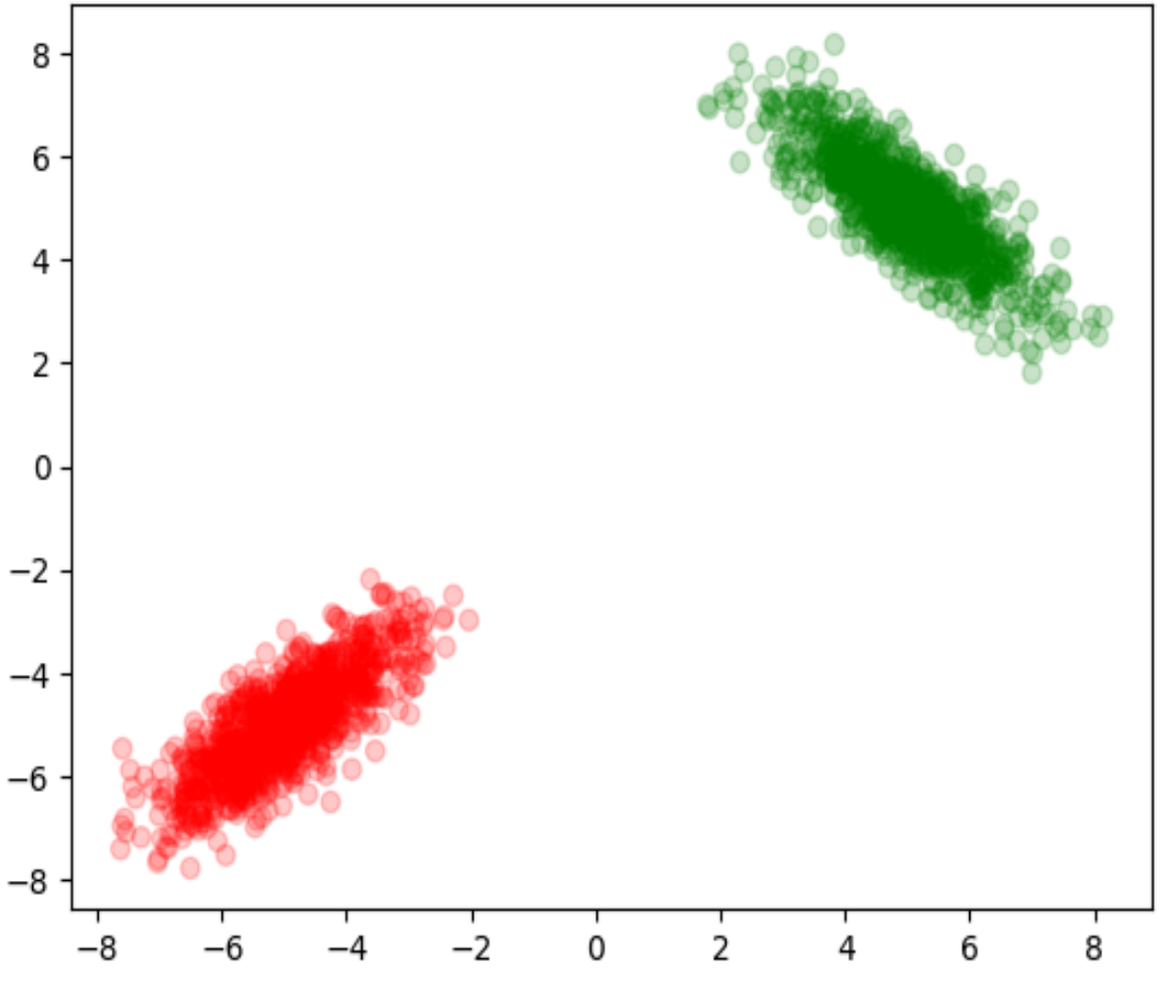
$$= \log\left(\frac{p}{1-p}\right) + d \cdot \log\left[\frac{1-p_j^1}{1-p_j^0}\right] + \sum_{j=1}^{d} x_j \cdot \log\left[\frac{p_j^1\left(1-p_j^0\right)}{p_j^0\left(1-p_j^1\right)}\right]$$

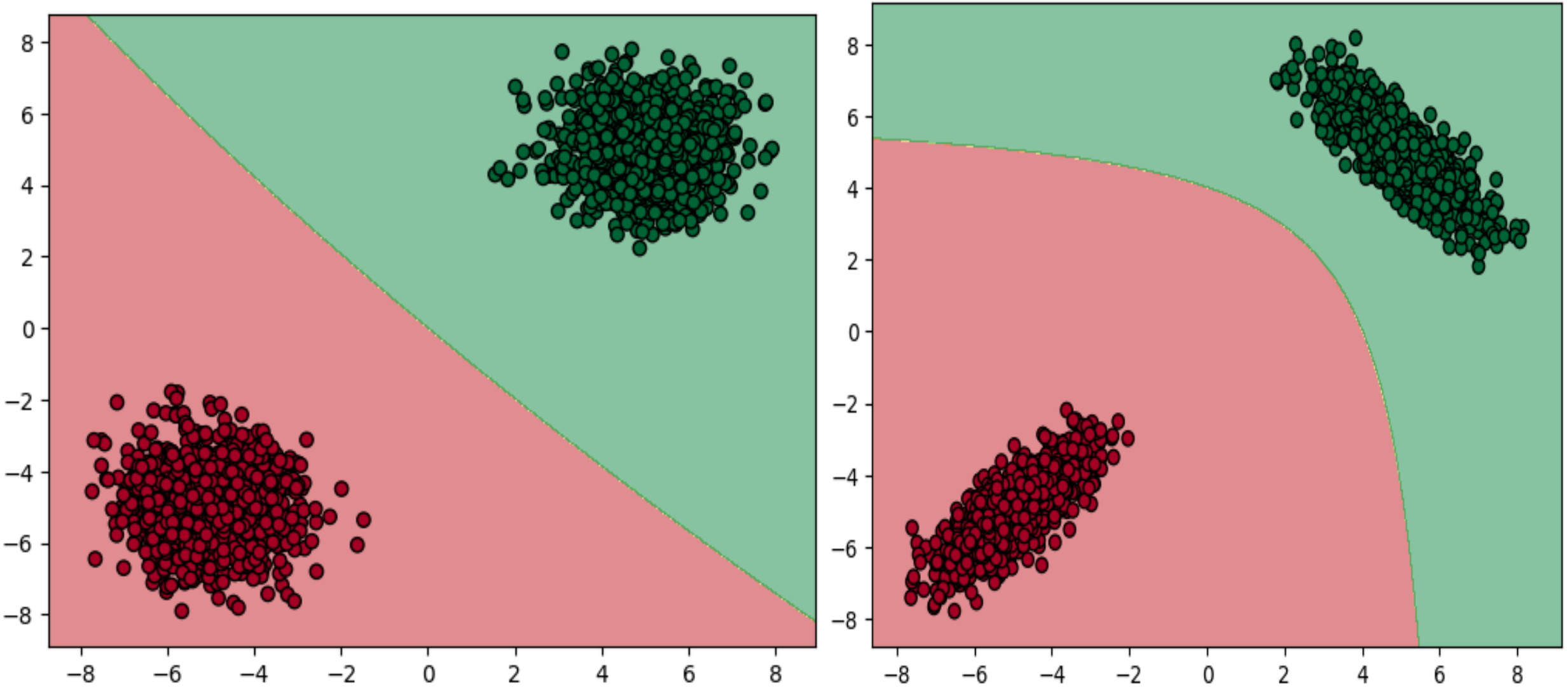
$$= b + \mathbf{w}^T\mathbf{x}$$

Linear/Quadratic Discriminant Analysis



Identical Covariance Matrices



Different Covariance Matrices

Linear/Quadratic Discriminant Analysis